

Chapter 37 — Infection and Inflammation of the Paediatric Genitourinary Tract

- **Lifetime childhood incidence:** about 1% of boys and 3–5% of girls have at least one urinary tract infection (UTI); **30–50% of these recur.**
- **Burden (United States):** about 1.1 million outpatient visits and 50,000 hospitalisations each year, with hospital charges above \$520 million.
- **Paradigm shift:** UTI has been **over-credited for scarring**; earlier guidelines (American Academy of Pediatrics [AAP] 1999 and others) assumed damage came from UTI alone and overlooked congenital dysplasia.

THE GOVERNING CONTRAST — OLD VERSUS CURRENT PARADIGM

	Earlier guidelines	Current understanding
Cause of scarring, hypertension, chronic kidney disease	Recurrent febrile UTI, especially with vesicoureteral reflux (VUR)	Congenital anomalies and renal dysplasia play the larger role
Strategy	Aggressive treatment and imaging to find VUR and scars	Selective imaging ; prompt treatment; prophylaxis for the high-risk child

Concept — *The whole chapter is written against the older assumption that every UTI threatens the kidney; each later section (imaging, prophylaxis, long-term sequelae) is a **retreat from blanket intervention towards risk-targeted care.***

EVALUATION AND MANAGEMENT OF A CHILD WITH A FEVER

- **Clinically significant fever:** rectal temperature $\geq 38^\circ\text{C}$ (100.4°F); in a previously healthy child aged **3–36 months**, $\geq 38^\circ\text{C}$ warrants evaluation.
- **Most fevers are viral**; 7–13% of children with no clear source have **occult bacteraemia or serious bacterial infection** (bacteraemia, bacterial gastroenteritis, cellulitis, meningitis, osteomyelitis, pneumonia, septic arthritis, UTI).
- **Age gradient:** serious bacterial infection is commoner **below 90 days** and especially **below 29 days**.
- **Goal:** do not miss serious infection; **start treatment quickly.**
- **“Toxic” appearance** carries a higher rate of serious infection: cyanosis, decreased activity, hyper- or hypoventilation, inability to interact with parents, irritability, lethargy, poor tone, poor perfusion, tachycardia, poor eye contact.
- **Predictors of bacteraemia** — toxic appearance, age **<30 days**, rectal temperature $\geq 39.4^\circ\text{C}$ — are strongly predictive, but **none guarantees** detection and **their absence does not exclude** serious infection.
- **UTI is the commonest bacterial infection in children:** about 7% of febrile infants and 7.8% of febrile children older than 24 months.

SYMPTOMS AND SIGNS PREDICTING UTI IN THE FEBRILE CHILD, BY AGE

Birth to 24 months	Older than 24 months (verbal)
Useful: fever $>40^\circ\text{C}$, previous UTI, suprapubic tenderness, uncircumcised penis Not useful (neither sensitive nor specific): vomiting, diarrhoea, poor feeding, irritability	Classic symptoms predictive: abdominal pain, back pain, dysuria, frequency, new-onset incontinence

KEY POINTS: DIAGNOSIS OF URINARY TRACT INFECTION IN A CHILD

- Signs and symptoms in young children **may be nonspecific**.
- UTI is the **commonest bacterial infection** in a child presenting with fever.
- A child who appears “**toxic**” needs close evaluation: serious infection is likely.

Definition of a Urinary Tract Infection

- **What counts as a “significant” UTI is controversial**; historically bacteria in urine defined infection because urine was thought sterile.
- **Urine is not sterile**: a **urinary microbiome** exists from as early as **2 weeks of age**, revealed by **expanded quantitative urine culture (EQUC)** and **16S ribosomal RNA gene sequencing**; its role is not yet understood.
- **AAP, age 2–24 months**: diagnosis must rest on urine from **suprapubic aspiration (SPA)** or **catheterisation**.
- **Bag specimen pathway**: bag urine for urinalysis only → if suggestive (positive leukocyte esterase or nitrite, or leucocytes or bacteria on microscopy) → **culture must come from SPA or catheter**.
- **Reason**: a positive culture from a bagged specimen has a **false-positive rate of up to 75%**.

Pathogenesis of Urinary Tract Infection Development in Children

- **Host, bacterial and immune factors** interact and are in **constant flux** in children — immunity and gut colonisation of a newborn differ completely from those of a 6-month-old.

Figure 37.1 — the ascending sequence: uncomplicated versus complicated UTI (Flores-Mireles et al., Nature Reviews Microbiology 2015)

Uncomplicated (A)	Complicated (B)
Gut uropathogen colonises urethra (1) → migrates to bladder (2) → colonises and invades umbrella cells via pili and adhesins (3) → innate inflammatory clearance begins (4) → evaders multiply (5) → biofilm (6) → toxins and proteases damage host cells (7) → ascent to kidneys (8) → renal colonisation (9) → tissue damage (10) → bacteraemia by crossing the tubular epithelial barrier (11)	Same steps 1–2, but infection requires a compromised bladder — commonest cause an indwelling catheter → catheter-induced inflammation with fibrinogen accumulation on the catheter (3) → attachment of uropathogens expressing fibrinogen-binding proteins → neutrophil infiltration (4) → multiplication (5), biofilm (6), epithelial damage (7) → renal seeding (8–9), tissue damage (10), bacteraemia (11)

Bacterial Factors Leading to Paediatric Urinary Tract Infections

Commensal bacteria	Virulent (uropathogenic) bacteria
Lack virulence traits to subvert host defences; can still cause UTI	Possess adaptations and fitness factors that subvert or hijack host defences and let them live where they normally would not; cause the majority of UTIs

- **Virulence** (Latin *virulentus*, “full of poison”): the ability of an organism to cause disease in a host.
- **Sequence of virulence**: attach to urogenital mucosa → trigger signalling and immunological cascades → **invade the bladder**.
- **Uropathogenic *Escherichia coli* (UPEC)** is the most studied and by far the commonest pathogen; **most *E. coli* UTIs are caused by UPEC** rather than commensal strains — evidence that virulence factors matter.
- **Four classes of virulence factor**: **adherence** to uroepithelium; **nourishment** in adverse environments; **protection** from host immunity; **toxins** enabling invasion.

Figure 37.2 — *E. coli* features relevant to pathogenicity (Johnson, *Clinical Microbiology Reviews* 1991)

- **Surface:** fimbriae, flagellum, capsule (K antigen), lipopolysaccharide (LPS; endotoxin, O antigen) on the outer membrane.
- **Secreted and membrane-bound:** exotoxins (α -haemolysin), membrane-bound toxins, siderophore-iron complexes; membrane proteins for transport and serum resistance.

UROPATHOGENIC E. COLI VIRULENCE FACTORS COMPARED

Factor	Class	Mechanism	Clinical association
Type 1 fimbriae	Adherence	FimH tip binds uroplakins ; triggers entry	Cystitis , asymptomatic bacteriuria
P fimbriae	Adherence	Bind Galα1–4Galβ digalactoside	Pyelonephritis
Aerobactin	Nourishment	Siderophore strips iron from host proteins	Growth in low-iron serum and dilute urine
α -Haemolysin	Toxin	Lyses bladder and renal cells; cytokine release	Inflammation
Cytotoxic necrotising factor 1	Toxin	Cytoskeletal disruption; aids binding and invasion	Invasion
Capsule (K antigen)	Immune evasion	Resists phagocytosis; anticomplementary	Pyelonephritis rather than cystitis
Biofilm	Persistence	Barrier to immunity and antibiotics	Recurrence
Flagella	Motility	Escape and nutrient seeking	Ascent bladder → kidney

- **Pattern:** type 1 fimbriae mark bladder strains; P fimbriae, capsule and flagella mark upper-tract strains.

Bacterial Fimbriae

- **Adherence is the first step** of UTI pathogenesis and the best-studied virulence trait; mediated by **adhesins** (pili, F antigens) — filamentous appendages projecting from the cell.
- **Chaperone-usher pathway (CUP) pili** are expressed by many gram-negative pathogens (*E. coli*, *Klebsiella*, *Proteus*, *Pseudomonas*, *Haemophilus*, *Salmonella*, *Yersinia*); **a single UPEC strain can encode many distinct CUP pili**.
- **Type 1 and P pili** are the best characterised; **type 1 pili are essential for bladder colonisation and invasion**.

	Type 1 pili	P (pyelonephritis-associated) pili
Mannose	Mannose-sensitive (the commoner class); blocked by D-mannose and concanavalin A	Mannose-resistant
Structure / receptor	Major rod subunit FimA ; minor subunits FimC, D, F, G, H; FimH tip adhesin binds uroplakins on umbrella cells → stimulates bacterial entry; older studies placed receptors in bladder muscle layers rather than epithelium	Agglutinate P blood group erythrocytes; receptor Galα1–4Galβ digalactoside in neutral glycosphingolipids of epithelium and red cells; receptor phenotypes common in the population; binding sites in kidney and bladder
Clinical	Strains with mannose-sensitive adhesins alone → cystitis and asymptomatic bacteriuria , not pyelonephritis	Found in most strains causing pyelonephritis
Step 1 — adhesion, signalling, activation		Step 2 — effector phase
Bacterial attachment to epithelial receptors → innate activation . Without step 1: host not alerted, mucosa inert → asymptomatic carrier state		Epithelial cytokines recruit inflammatory cells that kill bacteria. Without step 2: exaggerated inflammation with tissue destruction

Figure 37.3 — the two steps of pathogenesis (Bergsten, Wullt and Svanborg, *International Journal of Medical Microbiology* 2005)

- **Step 1 signalling is Toll-like receptor 4 (TLR4)-dependent** → cytokine secretion and receptor expression.
- **P and type 1 fimbriae use different receptors and different TLR4 pathways:** P fimbriae bind Gal α 1–4Gal β on glycosphingolipids (ceramide-anchored); type 1 fimbriae bind **α -mannose residues on glycoproteins** and also signal through TLR4.
- **Intracellular cycle:** adherence → signal transduction → **uptake into umbrella cells** → **intracellular bacterial communities (IBCs)**: growth slows until the IBC fills the cytoplasm → bacteria **flux out** and spread → re-enter epithelium → **quiescent intracellular reservoir (QIR)** in underlying transitional cells.
- **QIR:** a **handful of non-replicating bacteria**, viable for **months**; reactivated during urothelial turnover, they **seed the lumen and reinitiate UTI**.
- **TLR4-dependent expulsion** normally defends the urothelium: LPS on TLR4 → adenylyl cyclase → **cyclic adenosine monophosphate** → exocytic expulsion of UPEC from umbrella cells.
- **Intracellular invasion lets UPEC evade host defences and some antibiotics.**

Figure 37.4 — intracellular bacterial communities versus biofilm on inert surfaces (Kau, Hunstad and Hultgren, *Current Opinion in Microbiology* 2005)

UPEC on urothelium (A)	Biofilm on inert surface, e.g. catheter (B)
Colonisation and invasion → early IBC, rapid intracellular growth → mid IBC, biofilm-like properties → late IBC, fluxing out and filamentation → reservoir and recurrence	Attachment and extracellular polymeric substance production → early development → maturation → dispersion

Concept — The intracellular bacterial community is **a biofilm built inside a host cell**; that shared architecture explains why both catheter infections and recurrent cystitis resist clearance.

Aerobactin

- **Iron is essential** to *E. coli* (oxygen storage and transport, DNA synthesis, electron transport, peroxide metabolism).
- **Host withholds iron during infection:** reduced intestinal absorption, more iron-binding proteins, shift of plasma iron into intracellular stores.
- **Aerobactin**, a **siderophore**, extracts iron from host binding proteins and delivers it to bacterial iron centres.

Toxins

- **UPEC toxins:** **α -haemolysin (hlyA)**, **cytotoxic necrotising factor 1 (CNF1)**, autotransporter toxins — they alter host-cell function or shape, arrest the cell cycle or lyse cells.
- **α -Haemolysin** is produced by about half of UPEC strains; **CNF1** by a minority.

Capsular Polysaccharide

- **More than 80 capsular types** in *E. coli*: linear polymers coating the cell, **hindering antigen detection**.
- **UPEC surface = LPS covered by capsular K antigens**; both help escape host immunity.
- **Encapsulated K strains** are less phagocytosed and **anticomplementary** → impaired clearance; **impairment proportional to the amount of polysaccharide**.

Biofilm Formation and Flagella

- **Biofilm scaffold:** extracellular DNA, exopolysaccharides, pili, flagella and other adhesins → multicellular community **protected from immunity and antimicrobials**, with a **physical barrier to antibiotic entry**.
- **Biofilm-like IBCs** inside urothelial cells protect UPEC from neutrophils, antibiotics and stress.
- **Flagella** also act at several stages of **biofilm development**.

KEY POINTS: BACTERIAL FACTORS

- UTI bacteria are **commensal or virulent**; both cause UTI, but **virulent organisms cause the majority**.
- Virulent bacteria carry adaptations that **subvert or hijack host defences** and let them live where they normally would not.
- Virulence factors improve **adherence**, allow **nourishment**, **protect against immunity** and supply **toxins for invasion**.

Concept — Each virulence factor solves one obstacle to infection — **attach, feed, hide, damage, persist, ascend** — and the factors a strain carries predict whether it stays in the bladder or reaches the kidney.

Host Risk Factors Leading to Paediatric Urinary Tract Infections

- **Host determinants:** age, genetics, sex, race, and anatomical, functional and behavioural features.

Sex and Age

- **Boys exceed girls only below 1 year of age**; at every other age, including old age, **females predominate**.
- **Febrile children:** UTI incidence 3–5%; **girls carry the higher risk after 2 months**; at 2 months to 2 years, a febrile girl has about **2.3 times** the risk of a febrile boy.

Age band	Boys	Girls
First year of life	3.7% (higher)	2%
After the first year (incidence)	0.03–0.2%	3%
Cumulative by age 6 years	2%	7%
Older children and sexually active teenagers	1%	3%

Race

- **UTI is commoner in White girls** than in girls of other races.
- **Non-Black children** carry a **several-fold higher risk** of UTI and of recurrent UTI than Black children in meta-analysis, most marked in studies at low risk of bias.

Genetics

- **No specific gene identified**, but a genetic component is suggested by: **UTI persisting after VUR resolves**; more bacteriuria in **sisters** of children with recurrent UTI; **age at first infection** and a **mother with UTI** as the two greatest risk factors for recurrent UTI in women aged 18–30; **familial clustering of pyelonephritis** in pedigree studies.

Blood group system	Association
P1 (P fimbrial receptor)	Over-represented in girls with recurrent pyelonephritis
Lewis	Women with Le(a-b-) and Le(a+b+) have three times the recurrent UTI risk of Le(a-b+); Le(a-b-) also increased in children with UTI
Secretor (ABH antigens)	Non-secretor phenotype associated with renal scars on dimercaptosuccinic acid scan in children with VUR and UTI

Circumcision

- **Circumcision reduces UTI in the first 6 months of life by almost tenfold**, matching a period of heavy uropathogen colonisation of the **prepuce** that declines and resolves by **5 years**.
- **Wiswell and Roscelli 1986** (landmark; more than 200,000 boys, United States Army hospitals 1974–1983; UTI proven by catheter or SPA): **circumcised 0.11%, uncircumcised 1.12%, girls 0.57%**.

- **AAP Task Force on Circumcision 2012: benefits of newborn circumcision outweigh risks;** routine circumcision of all boys not justified, but benefits justify **access for families who choose it and third-party payment.**
- **Benefit beyond infancy** remains debated; case-control data show reduction before and after 1 year; **the size of benefit depends on baseline risk.**
- **Antenatal hydronephrosis** (meta-analysis): circumcision **protective**, including in the **VUR subgroup.**

Group (meta-analysis)	UTI risk	Number needed to circumcise to prevent one UTI
Normal healthy boys	0.5–1%	111
Recurrent UTI	10% recurrence	11
High-grade VUR	30% recurrence	4

Faecal and Perineal Bacterial Colonisation

- **Faecal–perineal–urethral route:** most UTI bacteria colonise gut, perineum and periurethral area, then ascend retrogradely.
- **Recurrent UTI in women:** most reinfections are the **same UPEC strain persisting in faecal flora for ≥12 months**, cycling through faecal and vaginal reservoirs.
- **Loss of protective vaginal lactobacilli** → enteric colonisation of vagina and periurethra → more *E. coli*; antibiotics, especially trimethoprim–sulfamethoxazole (TMP-SMX), eradicate lactobacilli.
- **Children:** mechanism unproven, but **antibiotic treatment of schoolgirls for non-urological infections (usually otitis media)** produced colonisation and bacteriuria with uropathogenic organisms.
- **Urinary microbiome** from 2 weeks of age (16S sequencing, EQUC): in **girls it changes through childhood until the postpubertal adult pattern; in boys it changes less.**
- **Presumed role:** symbiotic — metabolic and trophic functions and a **protective barrier against uropathogens**; in adult women the microbiome **differs in urge urinary incontinence**, implying a role in common urinary symptoms.

Anatomic Abnormalities

- **Malformation-associated UTIs usually appear before 5 years;** detection matters because many are **surgically correctable** and persistence causes renal damage and recurrence.
- **Abnormalities:** hydronephrosis, pelviureteric or vesicoureteric junction obstruction, VUR, infection stones, infected non-functioning renal segments, infected necrotic papillae.
- **AAP: renal and bladder ultrasound (RBUS) in every child aged 2–24 months after the first febrile UTI.**

Vesicoureteral Reflux

- **VUR: 1–2% of newborns, 25–40% of children after a first UTI.**
- **Most pyelonephritis occurs without VUR** (only a minority of scan-proven pyelonephritis has VUR); but kidneys with **grade III–IV reflux were twice as likely** to show pyelonephritic change.
- **Modifiers:** reflux grade, distal ureteric diameter, **early onset of reflux during filling**, age, sex, and **bladder and bowel dysfunction.** Detail in Chapter 41.

Sexual Activity

- **Sexually active girls have more UTIs;** UTI has been proposed as a **marker of teenage sexual activity**, but the relationship remains **poorly defined.**

Bladder and Bowel Dysfunction

- **Bladder and bowel dysfunction (BBD) = dysfunctional elimination syndrome:** infrequent voiding, incomplete emptying, constipation and/or bladder overactivity; contributes to **UTI and VUR.**

- **Presentations:** recurrent UTI (febrile or afebrile), day or night incontinence, urgency, frequency, constipation, encopresis.
- **Lower urinary tract dysfunction (LUTD)** affects about **20% of children**; among children with LUTD, **UTI is far commoner in girls than in boys**, mostly with urinary stasis, detrusor underutilisation and voiding dysfunction.

Overactive bladder	Dysfunctional voiding
Urgency with or without urge incontinence, usually with frequency and nocturia	Neurologically normal child with increased pelvic floor activity during voiding
Treatment: anticholinergics → VUR resolution or improvement in many	Treatment: biofeedback → VUR resolution in over half and grade improvement at 1 year

- **Treating BBD reduces recurrent UTI and improves VUR resolution.**
- **Constipation treatment:** relief of constipation → daytime incontinence largely disappears, enuresis often resolves, and **UTIs disappear in children without anatomical abnormality**; UTIs were commoner in constipated girls than boys.

Neurogenic Bladder

- **UTI is the commonest infection** in neurogenic bladder (spina bifida, spinal cord injury) — about **2.5 UTIs per year**; in spina bifida, **UTI is the commonest reason for emergency attendance**; brings higher morbidity, mortality and **end-stage renal disease**.
- **Cause is not only mechanical** (poor emptying, stasis, stones): a **dysfunctional bladder immune response** also contributes.
- **Clean intermittent catheterisation (CIC):** chronic bacteriuria and/or pyuria in **40–80%**, mostly asymptomatic and harmless.
- **Without VUR, asymptomatic bacteriuria (ASB) on CIC is not a significant risk for renal damage and needs no antibiotics.**
- **Stopping prophylaxis in spina bifida reduces bacterial resistance**; most children can undergo urodynamics without prophylactic antibiotics.
- **Daily prophylaxis with CIC** may delay bacteriuria briefly but gives **no long-term benefit and breeds resistance**.
- **Catheter type makes no difference:** sterile versus non-sterile, single-use versus reusable, lubricated versus non-lubricated.
- **In a catheterisation cohort**, risk factors are **high catheterisation volume in females** and **low catheterisation frequency in males**; prophylaxis gave **less bacteriuria but more clinical UTIs** → **timely, scheduled catheterisation** prevents UTI.

Figure 37.5 — multifactorial causes of UTI in neurogenic bladder (McKibben et al., *Urologic Clinics of North America* 2015)

- **Better uropathogen access:** perineal and urethral colonisation, altered protective flora, **direct inoculation** by catheterisation.
- **Compromised ability to fight infection:** immunodeficiency, **bladder ischaemia**.
- **Better adherence:** defective **glycosaminoglycan (GAG) layer**, impaired secretory immunoglobulin A response, defective apoptosis.
- **Compromised washout:** inefficient voiding and post-void residual urine, **hydrokinetic factors**, deficient mechanical cleansing by voiding.

Iatrogenic Factors

- **Catheter-associated UTI (CAUTI) is the commonest nosocomial infection** — more than 1 million cases yearly in United States hospitals and nursing homes.
- **Risk rises with catheter duration:** bacteriuria about **3–10% per day**.
- **Children:** nosocomial UTIs make up 6–18% of paediatric nosocomial infections; each adds about **one hospital day** and more than \$1000 per case.
- **Prevention works:** CAUTI incidence and mortality in catheterised adults **fell over 2001–2010** as prevention became a target.
- **Paediatric risk factors:** **duration of catheterisation**, contact precautions, prior catheterisation.
- **Prevention:** **judicious use and removal as soon as no longer necessary**; Healthcare Infection Control Practices Advisory Committee (HICPAC): catheters for **necessity, not convenience**; consider **external (condom) catheters** and **intermittent catheterisation**.

Immune Status

- **UTI in young healthy women is benign**; in **young children and the elderly** it is more complicated — **immunity is diminished at both ends of life**.
- **First few months of life = greatest risk:** serum immunoglobulin G **lowest at 1–3 months**; serum immunoglobulin A low and **absent or nearly absent on the urothelium** in early months.
- **Obesity:** inflammation, altered adipokine signalling, metabolic and epigenetic change → immune dysregulation; associated with UTI in **febrile children under 2 years** and with **acute pyelonephritis**.

KEY POINTS: HOST RISK FACTORS

- A multitude of **host factors** increase UTI susceptibility.
- UTI is commoner in boys than girls **only below 1 year**.
- **Circumcision reduces UTI in the first 6 months almost tenfold**.
- **Dysfunctional elimination syndrome** is important; treating bladder and bowel **reduces recurrence and improves VUR resolution**.
- **Without VUR, ASB on CIC** is not a significant risk for renal damage and needs no antibiotics.
- **CAUTI is the commonest nosocomial infection**; most are preventable by judicious catheter use.

Concept — Every host risk factor either **raises inoculum and exposure time** (prepuce, faecal colonisation, stasis, constipation, catheter) or **lowers defence** (immature immunity, obesity, receptor-bearing blood groups); prevention means reversing whichever applies.

CLASSIFICATION OF PAEDIATRIC URINARY TRACT INFECTIONS

- **Three axes:** complicated versus uncomplicated; upper (pyelonephritis) versus lower (cystitis); first versus recurrent.
- **Complicated UTI:** anatomical or functional abnormality, or a foreign body (ureteric stent, indwelling catheter); **neonatal and infant UTIs are presumed complicated** because anomalies are common and morbidity high.

Presumed pyelonephritis	Presumed cystitis
High fever , nausea, vomiting, flank pain, lethargy	Afebrile , lower tract symptoms only: urgency, frequency, dysuria, malodorous urine, suprapubic tenderness

- **Clinical localisation is not accurate: dimercaptosuccinic acid (DMSA) scan is the gold standard** for pyelonephritis; only **50–66%** of children with fever and systemic features show acute DMSA changes; a small subset of afebrile “lower” UTIs have acute pyelonephritis (APN) on DMSA.
- **Infants under 90 days** give hard-to-interpret signs: failure to thrive, diarrhoea, irritability, lethargy, malodorous urine, **asymptomatic jaundice**, oliguria, polyuria.

RECURRENT UTI — THREE MECHANISMS COMPARED

	Unresolved bacteriuria	Bacterial persistence	Reinfection
Urine sterile between episodes?	No	Yes	Yes
Cause	Inadequate therapy: non-compliance, malabsorption, suboptimal metabolism, resistant organism	Nidus not eradicated , shielded from antibiotics: anomaly, calculus, necrotic papilla, catheter, stent	New infection after cure
Organism	Same	Same organism each episode	Usually different
Remedy	Culture-directed retreatment	Find and remove the nidus	Prevention

Biofilms

- **Biofilm:** structured microbial community in a **self-made polymeric matrix**, adherent to a living or inert surface; drives **bacterial persistence**.
- **Why antibiotics fail:** (1) incomplete penetration; (2) **slow growth** resists agents needing active growth; (3) low expression of antimicrobial-binding proteins; (4) activated genes alter envelope, targets and susceptibility; (5) survival at **1000–1500 times** the usual killing concentration.
- **Renal biofilms:** bacteria adhere as thin biofilms to **urothelium and papillae** before invading parenchyma, and are less susceptible to antibiotics there.
- **Foreign bodies:** catheters, ureteric stents, calculi.

Extraluminal catheter colonisation	Intraluminal catheter colonisation
From gastrointestinal tract or perineum	From exogenous sources

- **Stents colonise far more often than urine shows bacteriuria** — a sterile urine does not mean a sterile stent.

Asymptomatic Bacteriuria

- **Kass definition:** two consecutive cultures $\geq 10^5$ colony-forming units (CFU)/mL of the same uropathogen in a patient free of infective symptoms.
- **E. coli is the commonest organism**, as in symptomatic UTI; likely **less virulent strains** → colonisation rather than infection.
- **“Asymptomatic” can mislead:** some women later became symptomatic and needed treatment, and most schoolgirls with ASB admitted **lower urinary tract symptoms and incontinence on careful questioning**.
- **Screening is not cost-effective:** given test performance and prevalence, screening yields many false positives, and treatment prevents no long-term harm.

Age group	Prevalence	Action
Full-term infants	<1%	Treat and obtain RBUS — may be the only sign of infection and a marker of underlying genitourinary disease ; consider voiding cystourethrogram (VCUG) in the premature infant with UTI and abnormal ultrasound (VUR incidence rises with each month of age)
Premature infants	3%	
Preschool	Girls 0.8%; boys fewer	No antibiotics if no VUR or genitourinary anomaly — no risk of recurrence, renal damage or impaired growth
School-age girls	1–2% per year; about 5% by age 15	No routine treatment : spontaneous clearance is common but bacteriuria recurs; treatment rarely gives lasting cure; very low risk of renal decline or hypertension; prophylaxis alters microbiota and breeds resistance

Bacterial Nephritis

- **Acute bacterial nephritis**: infection spreads through the kidney as an increasingly **suppurative process** with heavier leucocytic infiltrate and focal necrosis.
- **Generalised form = acute bacterial nephritis**; **localised form = acute focal bacterial nephritis (AFBN)**, **focal pyelonephritis**, **lobar nephronia**; clinical features of **septicaemia** often present.
- **Computed tomography (CT)**: global renal enlargement, perirenal fat inflammation, **thickened Gerota fascia**; on contrast, ill-defined, non-homogeneous, **wedge-shaped decreased enhancement** (Figure 37.6: CT of acute focal pyelonephritis, lobar nephronia).

Pyonephrosis

- **Purulent exudate from APN accumulating in a dilated collecting system**, typically a **hydronephrotic kidney with obstructed outflow**; CT: enlarged kidney, grossly dilated system with **higher-than-usual fluid attenuation**.

Acute Renal Abscess

- **Presents like pyelonephritis**, but **urine culture is negative in up to 20%** → diagnosis is **radiological**.
- **Ultrasound**: detects abscesses **≥1 cm** as sonolucent areas with low-amplitude echoes.
- **CT is most sensitive and specific**: early — well-defined low attenuation or decreased enhancement, or a **striated wedge-shaped zone**; later — coalescence into a **well-defined mass of homogeneous fluid attenuation** (pus).

SPECTRUM OF RENAL PARENCHYMAL INFECTION COMPARED

Entity	Defining feature	Imaging	Treatment
Acute focal bacterial nephritis	Localised suppurative nephritis	Wedge-shaped hypoenhancement; may mimic a mass	Antibiotics ≥3 weeks
Acute bacterial nephritis	Generalised form	Global enlargement, perirenal change	Antibiotics
Pyonephrosis	Pus in an obstructed dilated system	Dilated system, raised fluid attenuation	Drainage plus antibiotics
Renal abscess	Liquefied collection; culture negative in up to 20%	Well-defined mass of fluid attenuation	Antibiotics; drain if no response
Xanthogranulomatous pyelonephritis	Chronic, obstructive, <i>Proteus</i> or <i>E. coli</i>	Mass with perinephric extension, calculi	Nephrectomy (diffuse)

KEY POINTS: CLASSIFICATION

- UTIs may be classified in several ways.

- Pyelonephritis is usually diagnosed clinically (fever, flank pain, nausea, vomiting), yet **only 50–66%** of such children show acute DMSA changes.
- **ASB in 0.8% of preschool girls** (fewer boys); without VUR or anomaly they **need no antibiotics**.

Concept — In children *symptoms do not localise infection and bacteriuria does not equal infection*; classification therefore rests on imaging (DMSA) and on whether bacteriuria is causing harm, not on the culture alone.

DIAGNOSIS OF PAEDIATRIC URINARY TRACT INFECTION

- **No consensus definition:** international guidelines — AAP, National Institute for Health and Care Excellence (NICE), Italian Society of Pediatric Nephrology (ISPN), Royal Children's Hospital (RCH) Melbourne — **differ significantly**.
- **UTI = invasion of the urinary tract causing pathological change**, by the organism or by the host response.
- **Proof requires two things:** the organism in urine (**culture**) and evidence it is pathogenic (**symptoms, or an immune response in urine or blood**).

Symptoms

- **Classic adult symptoms** (dysuria, frequency, urgency, suprapubic or flank pain) become **harder to identify with decreasing age**.
- **Infants:** fever, irritability, poor feeding, jaundice, failure to thrive, vomiting, diarrhoea, abdominal distension, foul-smelling urine.
- **After the neonatal period fever is the primary symptom:** UTI prevalence about **5%** in febrile infants without another source; fever $\geq 38^\circ\text{C}$ for more than 2 days without source has a **positive likelihood ratio of 3.6** for occult UTI, rising with duration and height of fever.
- **Fever does not localise:** cystitis can be febrile.
- **Older children:** dysuria, incontinence, changed voiding, enuresis, flank or abdominal pain — still **nonspecific** (UTI in only a minority of such children); mimics include **BBD and vulvovaginitis**.
- **High prevalence means UTI must stay on the differential;** clinical probability tools combine symptoms, signs and laboratory results to decide when to test.
- **Risk factors to elicit:** **previous UTI** (children under 6 years recur at about **12% per year**), genitourinary anomaly, abnormal prenatal or postnatal ultrasound, family history, prior genitourinary or gastrointestinal surgery.
- **Adolescent urethritis:** consider **sexually transmitted infection** — *Neisseria gonorrhoeae*, *Chlamydia trachomatis*, *Ureaplasma urealyticum* — as well as routine uropathogens.
- **DUTY study** (Diagnosis of Urinary Tract infection in Young children; United Kingdom primary care, children under 5): predictors for **collecting urine** — **pain or crying on voiding, smelly urine, previous UTI, absence of severe cough, impression of severe illness**; adding **dipstick leucocytes, nitrite and/or blood** predicted need for antibiotics; the derived **points score outperformed clinical impression**.

Physical Examination

- **Specific findings are rare** in young children (fever, lethargy).
- **Another source does not exclude UTI:** an alternative focus (otitis, upper respiratory infection, gastroenteritis) only about **halves** the UTI risk; clinicians wrongly attributed most febrile UTIs to another source.

- **Classic history:** recurrent “otitis” before the first diagnosed UTI, then **DMSA scars** — earlier pyelonephritis misdiagnosed.
- **Abdomen:** palpable bladder (distension) or **flank mass (hydronephrosis)**; suprapubic, abdominal or flank tenderness; **costovertebral angle tenderness suggests pyelonephritis**.
- **Genitalia:** trauma, irritation, meatal stenosis, discharge, phimosis, foreign body, anomalies; boys — **testicular tenderness (epididymo-orchitis)**; girls — introital discharge, irritation, **ectopic ureter**, **prolapsing ureterocele**.
- **Back and neurology:** prominent fat pad, asymmetric gluteal cleft, sacral dimple (**spina bifida occulta**) plus neurological examination → neurogenic bladder dysfunction.

Laboratories

Urine Collection Methods

- **Collect urine before antibiotics** unless the delay would compromise a seriously ill child (NICE).
- **Earlier AAP guidance required pyuria** to separate true UTI from ASB or contamination.
- **Contamination rises as invasiveness falls.**

URINE COLLECTION METHODS COMPARED

Method	Contamination	Performance and role
Bag	50–60% (perineal and rectal flora, extraneous leucocytes)	Reliable only when normal ; never for culture
Clean catch (midstream)	About 26%	Against SPA: sensitivity 75–100%, specificity 57–100%; lower sensitivity risks missed or delayed diagnosis ; more reliable in older girls, circumcised boys, boys who retract the foreskin; preferred by NICE, ISPN, RCH when possible
Catheterisation	About 10–15%	Against SPA: sensitivity 95%, specificity 99%; recommended in non-toilet-trained children under 2
Suprapubic aspiration	<1% — reference standard	Invasive; trauma or infection; not always feasible in primary care

- **Catheterisation technique:** sterile technique → **intraurethral and/or topical lidocaine** (reduces discomfort) → in girls, **two-person technique:** each labium majus drawn gently **outward and slightly laterally** to expose the recessed meatus (excessive lateral pull hurts and makes the attempt traumatic) → insert → **discard the first drops** (urethral contamination).
- **SPA technique:** skin preparation → **22-gauge needle 1–2 cm above the pubic bone** → aspirate into a sterile syringe; blind success **45–65%**, rising to **80–90% with ultrasound** confirmation and guidance; local anaesthetic optional, benefit unproven.

Urinalysis

- **Urinalysis (UA)** = dipstick plus microscopy; examine within **1 hour at room temperature or 4 hours refrigerated**.
- **Culture is the gold standard** but needs **≥18 hours** for growth and **2–3 days** for the final result and susceptibilities; UA gives a **rapid probability** to guide empirical treatment.
- **Urine concentration** alters the likelihood ratios of dipstick and microscopy results, although thresholds are not adjusted for specific gravity.

Urine dipstick tests

- **Leukocyte esterase (LE):** released from lysed leucocytes → **marker of pyuria**; **sensitive but less specific**; false positives from other inflammation; **2+ raises probability more than 1+**.

- **Pyuria and LE are commoner with *E. coli*** than with *Klebsiella* or *Enterococcus* → **culture a suspected UTI even without pyuria.**
- **LE misses a fraction of UTIs**, but its absence helps separate ASB from true UTI because **true UTI without pyuria is rare.**
- **Nitrite:** dietary nitrate reduced by **gram-negative enterics** over **several hours** → **first-morning urine** is best.
- **Nitrite false negatives:** **frequent voiding in infants**, dilute urine, **gram-positive organisms**; hence **low sensitivity but very high specificity** — a positive nitrite is very likely a true UTI.

RAPID URINE TESTS COMPARED

Test	Sensitivity	Specificity	Use
Leukocyte esterase	~79%	~87%	Negative helps rule out
Nitrite	~50%	~98%	Positive rules in
LE or nitrite positive	88%	79%	Screening
LE or nitrite or microscopy positive	99.8%	70%	Best rule-out combination
LE and nitrite both negative	Negative likelihood ratio 0.22		UTI much less likely, but continue antibiotics if clinical picture demands
LE, nitrite, blood, protein all negative	Rule-out sensitivity about 97%		Four-parameter rule-out
LE + nitrite + blood	Specificity about 97%		Best rule-in combination
Microscopic pyuria ≥5 white cells/high-power field	~74%	~86%	No advantage over LE
Bacteria, uncentrifuged fresh urine	~88% (unstained), ~91% (Gram)	92–96%	Most reliable rapid test

Urine microscopic examination

- **Pyuria threshold:** traditionally **5 white blood cells per high-power field (WBC/HPF)** on centrifuged urine; **10 WBC/HPF** is more reliable under 2 years (NICE, ISPN, RCH) with fewer false positives; **20–50 WBC/HPF or more** markedly raises post-test probability.
- **Microscopy for leucocytes adds nothing over LE;** its additive value to the dipstick is questionable.
- **Microscopic bacteria on uncentrifuged fresh urine = most reliable rapid test;** Gram stain improves accuracy but **routine Gram stain does not enhance diagnosis** in children having UA.

Urine Culture

- **A positive culture is essential** for diagnosis; culture is generally **unnecessary if UA is normal** (negative nitrite and LE, no pyuria or bacteriuria).
- **Threshold trade-off:** lower threshold → false positives and overtreatment; higher → undertreatment.
- **Historical Kass threshold $\geq 10^5$ CFU/mL** came from **morning-voided urine of adult women** and may not suit children; **frequent voiding and high output lower counts.**

CULTURE THRESHOLDS BY GUIDELINE BODY

Body	Specimen	Threshold
Historical (Kass)	Voided	$\geq 10^5$ CFU/mL
AAP 2011 (2–24 months)	Catheter or SPA	$\geq 50,000$ CFU/mL plus pyuria on UA
European Association of Urology	Clean catch, midstream or catheter	10^3 – 10^4 CFU/mL if a monoculture
RCH Melbourne	SPA / catheter	Any gram-negative growth / $> 10^3$ CFU/mL

Body	Specimen	Threshold
ISPN	Catheter / midstream	$>10^4$ / $>10^5$ CFU/mL

- **Pattern:** the more invasive the specimen, the lower the accepted count; the trap is applying the adult 10^5 figure to a catheter or SPA sample.
- **Contamination is more likely** with **low colony counts**, **heavy mixed growth** (but a true UTI plus contamination remains possible) and **non-pathogens:** *Lactobacillus*, coagulase-negative staphylococci, *Corynebacterium*, α -haemolytic streptococci, *Candida*.

Serum Tests

- **Renal invasion** → Toll-like receptor signalling → cytokines and chemokines; **urinary interleukin (IL)-6, IL-8, transforming growth factor- β 1** may rise.
- **Systemic response:** fever, white cell count, **erythrocyte sedimentation rate (ESR)**, **C-reactive protein (CRP)**, **procalcitonin (PCT)**.
- **Renal biomarkers combined** — neutrophil gelatinase-associated lipocalin (NGAL), kidney injury molecule-1 (KIM-1), urinary creatinine — improve prediction of **renal scars in children with VUR**.

Marker	Localises upper tract?
Fever, ESR, peripheral white count	No — do not reliably separate upper from lower UTI
CRP	Reasonable sensitivity, low specificity
Procalcitonin (rises with bacterial endotoxin)	More robust than CRP or white count for early APN and later DMSA scarring (meta-analysis); cut-off ≥ 0.5 ng/mL, accuracy only moderate — best combined with other data to guide imaging

KEY POINTS: SYMPTOMS AND LABORATORY ANALYSIS

- Symptoms in infants and young children are **typically nonspecific**.
- **Another source of fever does not exclude a concurrent UTI.**
- **Bag urine is reliable only when normal** because contamination is frequent.
- Diagnostic criteria for UTI **vary between guidelines**.
- **LE: high sensitivity, lower specificity; nitrite: high specificity, lower sensitivity.**

Concept — Every step of diagnosis is a **sensitivity-specificity trade**: collection method trades invasiveness for contamination, LE and nitrite trade in opposite directions, and culture thresholds fall as the specimen becomes cleaner.

Radiographic Imaging

Controversies With Imaging Strategies

- **No evidence that routine imaging prevents** scarring, hypertension or renal failure.
- **Traditional work-up** = ultrasound plus VCUG; VUR is found in **more than 30%** of febrile UTIs, but **prophylaxis is unproven in low-grade VUR** → routine VCUG after every first UTI questioned.
- **NICE:** ultrasound routinely **under 6 months**; above 6 months only for **recurrent or atypical UTI** (atypical defined under Table 37.1).
- **AAP 2011 guideline is now defunct.**
- **AAP Section on Urology:** VCUG remains an **accepted option** — children with higher-grade VUR benefit from early detection and treatment; the **Randomized Intervention for Vesicoureteral Reflux (RIVUR)** trial showed prophylaxis **halved febrile or symptomatic UTI over 2 years**.

- **Yield:** NICE and AAP protocols would **miss about half or more of grade III–V VUR and most or all renal scars**; NICE performs worse in **boys** (negative predictive value).
- **Counter-argument:** most children without high-grade VUR are **spared the trauma, radiation and cost** of VCUG and DMSA; high-grade VUR declares itself at the **recurrent UTI**, and prompt treatment of that recurrence keeps renal risk low.
- **Top-down approach (TDA):** **DMSA after the first febrile UTI, VCUG only if DMSA abnormal** → misses only 15–30% of dilating VUR, but **lower specificity**.
- **Late DMSA:** NICE — **4–6 months after acute infection** for children **under 3 with atypical or recurrent UTI** and **over 3 with recurrent UTI**; AAP — no routine DMSA.
- **Bottom line:** **no strategy tests only the children who will benefit**; if a result will not alter management, the test should not be ordered.

TABLE 37.1 SUMMARY OF THE FIVE IMAGING RECOMMENDATIONS

Guideline	Ultrasound	Voiding cystourethrogram	Late DMSA
RCH Melbourne	If <3 months or seriously unwell	No	No
NICE <6 months	Yes	Atypical / recurrent*	Atypical / recurrent
NICE >6 months	Atypical / recurrent*	No	Atypical / recurrent*
TDA	No	If acute DMSA positive	If acute DMSA positive
AAP	Yes	If ultrasound positive	No
ISPN <1 year	Yes	If ultrasound positive and/or risk factors†	Yes
ISPN 1–5 years	Yes	If ultrasound or DMSA positive	Yes
ISPN >5 years	Yes	If ultrasound positive	If ultrasound positive

* Atypical UTI: seriously ill, poor urine flow, abdominal or bladder mass, raised creatinine, septicaemia, failure to respond to correct antibiotic within 48 hours, non-*E. coli* infection.

† ISPN risk factors: abnormal prenatal urinary tract ultrasound, family history of VUR, septicaemia, renal failure, **male infant under 6 months**, likely family non-compliance, abnormal bladder emptying, no response to correct antibiotic within 72 hours, non-*E. coli* infection.

Data from La Scola et al., Pediatrics 2013; Zhu et al., BMJ Open 2022.

- **What the table is for:** the bodies agree on ultrasound (except TDA) and **disagree on what triggers VCUG** — **ultrasound-driven** (AAP, ISPN), **clinical-course-driven** (NICE: atypical or recurrent), **DMSA-driven** (TDA), or **never routinely** (RCH).
- **Traps:** the **TDA uses no ultrasound**; **RCH images least**; **NICE omits VCUG over 6 months**; ISPN is the **only body using age bands up to and beyond 5 years** and the most imaging-intensive.

Ultrasound

- **Non-invasive, radiation-free, widely available** → routine use widely accepted; usually the first study.
- **Yield after first febrile UTI:** abnormal in about **15%**; only **1–2%** need further evaluation or treatment.
- **Relative indications in older children:** recurrent febrile UTI, poor response to antibiotics, **hypertension**, family history of renal or urological disease.
- **Shows:** size, shape and presence of both kidneys; congenital anomalies, obstruction, hydronephrosis, stones, **pyonephrosis**, renal or perirenal **abscess** → obtain **urgently if unusually ill or not responding**.

- **Acute-phase pitfalls:** oedema **overestimates renal size**; lobar nephronia **mimics a mass or tumour**; *E. coli* endotoxin **dilates the renal pelvis**, mimicking hydronephrosis.
- **Relative renal volume correlates with relative function** on scintigraphy, but ultrasound **misses more than 10% of scars**; useful to select children for DMSA.
- **NICE:** no ultrasound at **6 months–3 years** after a first febrile UTI unless atypical; high-resolution ultrasound has improved detection of acute renal involvement.
- **Very low sensitivity for VUR, even high grade**; in children under 24 months ultrasound **missed most of those needing surgical or medical intervention**; a **normal ultrasound does not lower the risk of recurrent pyelonephritis**.

Voiding Cystourethrogram

- **Contrast (fluoroscopic) or radionuclide (^{99m}Tc -pertechnetate) cystography; contrast VCUG is the gold standard for detecting and grading VUR.**
- **Contrast VCUG first** (better anatomy); **radionuclide cystogram for follow-up**.
- **Added information:** bladder size, shape, **trabeculation, diverticula**; **voiding images (essential)** — sphincter function and **urethral obstruction**; stool pattern (**constipation**); spinal defect (**spina bifida occulta**, possible tethered cord).
- **Timing:** as soon as **urine is sterile, the child is asymptomatic and voiding normally**.
- **A negative VCUG does not exclude VUR:** about a third with previous VUR and one negative nuclear study were positive a year later.
- **Cyclic VCUG** (fill → void → refill → void) increases sensitivity for **VUR and ectopic ureter**.
- **AAP standard protocol** (Sections of Urology and Radiology): **gravity filling from 100 cm above the table, at least two fill-void cycles, and record the volume at onset of VUR** — an independent predictor of **resolution and breakthrough UTI**.
- **Contrast-enhanced voiding urosonography (ceVUS):** real-time, **radiation-free**, high sensitivity against VCUG; some guidelines advise it for **first evaluation of VUR in girls** and for **follow-up**.

^{99m}Tc -Dimercaptosuccinic Acid

- **DMSA with single-photon emission computed tomography (SPECT) = gold standard for renal parenchymal lesions.**
- **Mechanism:** intravenous → **bound to proximal tubular cells** → excreted very slowly → stable cortical images, acquired **2–4 hours** after injection; radiation about **1 millisievert (mSv)**; CT detects some APN lesions DMSA misses, but CT has major drawbacks in children.
- **Timing is everything:** sensitivity for APN is **maximal within 1 week** of symptoms and **falls markedly by 1 month**.
- **Acute DMSA:** cortical uptake defects; kidney may look enlarged (inflammation, oedema).
- **Scar assessment no earlier than 6 months after APN** (some wait 1–2 years for reversible defects to resolve); **most acute lesions are transient**, about **15% scar**.
- **Scars** = cortical areas of decreased uptake, **indistinguishable from congenital dysplasia** (often associated with VUR).
- **Abnormal DMSA** is commoner with **dilating VUR**; scarring is worse with higher grade; an abnormal DMSA **lowers spontaneous VUR resolution independent of grade**; **older children show more defects** (scars are cumulative).

- **^{99m}Tc -mercaptoacetyltriglycine (MAG3):** less cortical detail but images the **collecting system** (obstruction); reduces DMSA false positives in a **very hydronephrotic kidney**; faster excretion → shorter study and **lower dose to bladder and gonads**.

Computed Tomography

- **Detailed anatomy and excellent sensitivity** for renal infection, but **high radiation (10–15 mSv for a paediatric abdomen and pelvis)** and contrast risks limit use.
- **Exceptions: non-contrast CT for suspected urolithiasis not seen on ultrasound;** distinguishing inflammation from tumour when acute ultrasound suggests a mass.
- **Findings:** cortical hypoattenuation, wedge-shaped defects, **loss of corticomedullary differentiation**, striations; abscess and AFBN may look alike acutely → **delayed scans** may be needed.

Magnetic Resonance Imaging

- **Excellent anatomical and functional imaging**, limited by cost, **need for sedation or anaesthesia** and availability; gadolinium risks **nephrogenic systemic fibrosis** in impaired renal function.

IMAGING MODALITIES COMPARED

Modality	Radiation	Answers	Key limitation
Ultrasound	None	Anatomy, obstruction, collections	Poor for VUR; misses scars
Contrast VCUG	Fluoroscopic	VUR grade, bladder, urethra	Catheterisation; negative study does not exclude VUR
ceVUS / radionuclide cystogram	None / low	VUR detection, follow-up	Less anatomical detail
DMSA ($\pm$ SPECT)	~1 mSv	Acute APN, scars, relative function	Timing-dependent; scar $\approx$ dysplasia
MAG3	Lower than DMSA to bladder and gonads	Cortex plus drainage/obstruction	Less cortical detail
CT	10–15 mSv	Infection extent, stones, mass versus inflammation	Radiation, contrast
MRI	None	Anatomy and function	Sedation, cost; gadolinium and nephrogenic systemic fibrosis

KEY POINTS: RADIOGRAPHIC IMAGING

- Imaging guidelines conflict and **all fall short** of testing only the children who will benefit.
- **Ultrasound** is non-invasive, radiation-free and available, but **unreliable for VUR** and misses some scars.
- **VCUG** is the most reliable method to **detect and grade VUR** and image bladder and urethra.
- **DMSA** is the **gold standard for renal cortical imaging** and gives relative renal function.

Concept — Each study answers one question — **ultrasound anatomy, VCUG reflux and outlet, DMSA parenchyma** — and none substitutes for another; the guideline controversy is only about **which children need the second and third question asked**.

MANAGEMENT OF PAEDIATRIC URINARY TRACT INFECTION

Antibiotic Treatment

- **Goals:** eradicate infection, **prevent renal scarring**, relieve symptoms.

- **Urine is usually sterile within 24 hours** of effective therapy.
- **Each new febrile UTI raises scar incidence** (about 2.8% per episode).
- **Early treatment limits renal involvement and scarring:** starting antibiotics at day 3 rather than day 2 sharply increased acute scintigraphic lesions, and scar formation rose steeply as the start slipped towards day 6.
- **Anti-inflammatory steroids** (dexamethasone, methylprednisolone) to limit scarring: results vary — **investigational**.
- **Empirical start is routine** (nonspecific symptoms; culture takes 2–3 days) — ideally **based on UA and adjusted to culture**; in practice UA and culture are often omitted → **overtreatment** of non-UTIs or **undertreatment** of resistant UTIs; indiscriminate broad-spectrum use → side effects and resistance.

Inpatient Versus Outpatient Management

- **Decision factors:** likely uropathogen, clinical status, **family reliability**; **fewer than 1%** of outpatients evaluated for UTI need admission.
- **Outpatient:** infants **older than 2 months** and **non-toxic** children with suspected pyelonephritis, if oral antibiotics are tolerated and compliance assured.
- **Oral equals intravenous** for time to improvement and scar prevention in randomised trials — urinary antibiotic levels far exceed serum.
- **Newborns and young infants** → **admit and give parenteral antibiotics:** more **urosepsis**, **positive blood cultures** in about 20%, **hyponatraemia** and **hyperkalaemia**.
- **Admission indications:** **age under 1 month** (some say under 2 or even 6 months), **toxic appearance or dehydration**, **oral intolerance**, **doubtful compliance**.
- **Neonatal pathway:** admit → **full septic evaluation** → parenteral antibiotics → culture confirms and symptoms improve → switch to oral; **defervescence takes at least 24 hours**; **≤2 days intravenous then oral is safe** in young infants.
- **Most children (90%) are afebrile within 48 hours**; **not improving at 48 hours** → **RBUS** and, if culture not back, **broaden cover**.

Antibiotic Duration

Infection	Duration	Notes
Febrile UTI	7–14 days	Shorter courses proven inferior
Afebrile cystitis	2–4 days	Fewer recurrences than single-dose or 1-day; equivalent to 7–14 days
Focal pyelonephritis (lobar nephronia)	At least 3 weeks	Sufficient in most
Renal abscess	Antibiotics often suffice	Drain if no clinical response or resolution
Obstructive uropathy with UTI	—	Temporary tube or stent drainage

Antibiotic Selection

- **Gram stain**, if available, guides the initial empirical choice.
- ***E. coli* is the commonest paediatric uropathogen**; **TMP-SMX and amoxicillin** are used in about half of outpatient UTIs but are **poor empirical choices** (high *E. coli* resistance).
- **Nitrofurantoin or a first-generation cephalosporin** = appropriate **narrow-spectrum** choice for many, adjusted for age and comorbidity.

- **Susceptibility reports use serum breakpoints**, yet **urinary levels may exceed serum more than 100-fold** → narrow agents such as **cephalexin** may be wrongly excluded in favour of an **overly broad third-generation cephalosporin** (resistance, *Clostridioides difficile*).
- **Prevalence and resistance vary by sex, setting and prior antibiotics** → empirical therapy from **local or regional antibiograms**, revised annually.
- **Other uropathogens**: gram-negative — *Klebsiella*, *Proteus*, *Enterobacter*, *Citrobacter*; gram-positive — *Staphylococcus saprophyticus*, *Enterococcus*, rarely *S. aureus*.
- **Neonates and young infants: cover *Enterococcus*** (commoner in early infancy); it is usually **ampicillin-sensitive** and **intrinsically resistant to cephalosporins**.
- **Safe empirical parenteral regimen in neonates and young infants: ampicillin plus a third-generation cephalosporin or an aminoglycoside**.
- **Aminoglycosides**: once-daily dosing; consider when ***Pseudomonas* risk** (recent antibiotics, urinary tract abnormality).
- **Intravenous** → **oral** in most after several days, guided by culture and **defervescence**.
- **Empirical broad-spectrum cover** for children at risk of resistance: previous UTI, recent antibiotics, recent hospitalisation, genitourinary anomaly. Broad-spectrum = antipseudomonal and β -lactam/ β -lactamase-inhibitor penicillins, macrolides, fluoroquinolones, second- to fourth-generation cephalosporins, lincosamides, carbapenems.
- **Fluoroquinolones**: highly effective but resistance rising → **not first line**; reserve for resistant organisms such as *Pseudomonas aeruginosa*; toxicity (cartilage, tendinitis, tendon rupture, arthropathy) similar to adults; **United States Food and Drug Administration (FDA) restricts paediatric urological use to complicated UTI including pyelonephritis**.

TABLE 37.2 UROPATHOGEN PREVALENCE BY SEX AND CLINICAL SETTING

Organism	Boys — outpatient	Boys — inpatient	Girls — outpatient	Girls — inpatient
<i>Escherichia coli</i>	50%	37%	83%	64%
<i>Enterococcus</i>	17%	27%	5%	13%
<i>Klebsiella</i>	10%	12%	4%	10%
<i>Proteus mirabilis</i>	11%	5%	4%	2%
<i>Pseudomonas aeruginosa</i>	7%	10%	2%	6%
<i>Enterobacter</i>	5%	10%	1%	4%

National data, The Surveillance Network; prevalence varies by region. Data from Edlin et al., Journal of Urology 2013.

- **Pattern**: *E. coli* dominates **outpatient girls** and falls in **boys and inpatients**; *Enterococcus*, *Klebsiella*, *Pseudomonas* and *Enterobacter* rise in inpatients; *Proteus* is a boys' organism.
- **Trap**: "*E. coli* causes more than 80% of paediatric UTIs" is true only of **outpatient girls** — an inpatient boy has barely one chance in three.

TABLE 37.3 DETAILS OF COMMON ANTIBIOTIC DOSING

Agent	Dose	Common side effects	Restrictions / comments
Oral			
Amoxicillin-clavulanate	20–40 mg/kg/day in 3 doses	Diarrhoea, nausea, vomiting, rash	
TMP-SMX	6–12 mg/kg/day of TMP in 2 doses	Diarrhoea, nausea, vomiting, photosensitivity, rash	Contraindicated <6 weeks

Agent	Dose	Common side effects	Restrictions / comments
Cefixime	8 mg/kg/day in 1 dose	Abdominal pain, diarrhoea, flatulence, rash	
Cefpodoxime	10 mg/kg/day in 2 doses	Abdominal pain, diarrhoea, nausea, rash	
Cefprozil	30 mg/kg/day in 2 doses	Abdominal pain, diarrhoea, raised liver enzymes, nausea	
Cephalexin	50–100 mg/kg/day in 4 doses	Diarrhoea, headache, nausea, vomiting, rash	
Nitrofurantoin	3–5 mg/kg/day in 2 doses	Nausea, vomiting, bad taste	Contraindicated <3 months , glomerular filtration rate <50%, glucose-6-phosphate dehydrogenase (G6PD) deficiency
Parenteral			
Ceftriaxone	75 mg/kg/day in 1 dose		Single daily dosing acceptable
Cefotaxime	150 mg/kg/day divided 6–8-hourly		
Ceftazidime	100–150 mg/kg/day divided 8-hourly		Antipseudomonal
Gentamicin	7.5 mg/kg/day divided 8-hourly		Single daily dosing acceptable alternative
Tobramycin	5 mg/kg/day divided 8-hourly		
Piperacillin	300 mg/kg/day divided 6–8-hourly		Antipseudomonal

- **Pattern: once-daily agents** are ceftriaxone, cefixime and (as an alternative) gentamicin; the **age restrictions sit on TMP-SMX and nitrofurantoin** — the two cheapest agents.

AGE AND SAFETY RESTRICTIONS ON ANTIMICROBIALS

Agent	Restriction	Reason
Nitrofurantoin	Under 3 months ; G6PD deficiency; febrile UTI	Haemolytic anaemia; low serum and tissue levels
Trimethoprim / TMP-SMX	Premature infants and under 6 weeks	Sulfonamides displace bilirubin from albumin → kernicterus
Fluoroquinolones	FDA: complicated UTI and pyelonephritis only	Cartilage, tendon, joint toxicity; resistance
Fluconazole	Under 6 months (per chapter)	—

Concept — Choose the **narrowest agent the local antibiogram allows**, reading susceptibility in the light of **urinary, not serum, concentrations**; age decides both the setting (admit the neonate) and which drugs are excluded.

MANAGEMENT AFTER URINARY TRACT INFECTION

- **Aim: prevent recurrence.** Repeat culture is **unnecessary** after treatment chosen on susceptibilities.

- **10–30% develop at least one recurrence**; risk is **highest in the first 3–6 months**, and **the more recurrences, the more likely the next**.
- **Counsel parents** to seek prompt evaluation of later febrile illness — early treatment limits renal damage.
- **Monitor height, weight and blood pressure** after any febrile UTI.

Recurrence after first UTI	Index UTI <1 year		Index UTI >1 year		
Boys	18%		32%		
Girls	26%		40%		
Pyelonephritic episode	1st	2nd	3rd	4th	5th
Renal scarring risk	5%	10%	20%	40%	60%

- **Infection model**: UTI follows **increased inoculum and exposure time** (stasis, constipation) and **compromised local or systemic immunity** (infrequent voiding, retention, high pressures from dysfunctional voiding or obstruction); **surgical correction of obstruction** reduces stasis; a **recent UTI** itself predisposes; when risk factors cannot be found or modified, **non-specific prevention** is used.

Prophylactic Antibiotics

- **Efficacy questioned** by small randomised series including **low-grade VUR**.
- **Neither AAP nor NICE recommends routine prophylaxis** after a first UTI.
- **Benefit is demonstrable in high-risk groups**, e.g. **girls with dilating VUR (grade ≥III)** (Swedish Reflux Trial and other randomised trials).
- **Low-risk features** (benefit hard to show): **circumcised boy, no BBD, no recent UTI, normal ultrasound or DMSA, no anatomical abnormality, no dilating VUR**.
- **Resistance**: prior exposure makes later UTIs more likely to be **resistant to the same agent** (faecal flora become resistant) — roughly **threefold**; so **the prophylactic agent should differ from the treatment agent**.
- **Ideal prophylactic**: covers most uropathogens, easy and well tolerated, **high urinary and low serum levels**, minimal effect on flora and resistance.
- **Dose**: about **one-quarter of the therapeutic dose**, given **at bedtime** in toilet-trained children to prolong bladder exposure.
- **Unanswered**: which patients benefit, ideal dose and schedule, alternating agents; **non-compliance is common**.

LANDMARK TRIAL — RIVUR (RANDOMIZED INTERVENTION FOR VESICoureteral REFLUX)

Element	Detail
Design	Prospective; TMP-SMX prophylaxis versus placebo ; 607 children, grade I–IV VUR , mostly after a first UTI, median age 12 months; included BBD
Recurrent febrile or symptomatic UTI	Risk roughly halved (reported as a 50% reduction over 2 years); the chapter also gives 25.5% versus 37.4%, relative risk 0.68 — a different denominator or endpoint definition; the gap widened over time
Number needed to treat	8 children for 2 years to prevent one febrile or symptomatic UTI; 5 in high-risk versus 18 in low-risk children
Renal scarring	No significant difference
Greatest benefit	BBD at baseline, febrile index UTI, higher-grade VUR

PROPHYLACTIC AGENTS COMPARED

Agent	Strength	Limitation
TMP-SMX	Traditional choice	Rising <i>E. coli</i> resistance ; sulfonamides displace bilirubin → kernicterus — avoid first 6 weeks
Trimethoprim	Alternative to combination	Avoid in premature and under 6 weeks
Nitrofurantoin	Minimal effect on faecal flora; resistance low	Low serum levels → not for APN or urosepsis ; haemolysis in G6PD deficiency (about 10% of Black, Sardinian, non-Ashkenazi Jewish, Greek, Eti Turk and Thai populations; oxidises haemoglobin to methaemoglobin); rare pulmonary fibrosis with long-term use
First-generation cephalosporin	Narrow spectrum	—

Concept — Prophylaxis **prevents recurrences, not scars**, and its benefit scales with baseline risk while its cost (resistance) does not — hence prophylaxis for the high-risk child only.

NONANTIBIOTIC PROPHYLAXIS**NONANTIBIOTIC PREVENTION COMPARED**

Measure	Mechanism	Evidence	Verdict
Cranberry	Impedes adherence; alters virulence factors	Mixed-age meta-analysis: no benefit ; paediatric-only review: comparable to antibiotics, better than placebo ; no benefit in neurogenic bladder	Not recommended routinely ; tolerability limits
Probiotics (<i>Lactobacillus</i>)	Restore protective flora	No benefit as monotherapy ; modest benefit as adjuvant to antibiotics	Adjunct only
D-mannose	Mimics receptor → competitively blocks type 1 fimbrial attachment	Fewer recurrences than observation and fewer side effects than nitrofurantoin — adult women only , limited quality	Unproven in children
Circumcision	Removes preputial colonisation	Reduces UTI in infants and toddlers; greatest in recurrent UTI and dilating VUR ; protective in antenatal hydronephrosis	Considered a medical indication in high-risk boys — controversial
Topical steroid for physiological phimosis	Allows retraction and hygiene	Significantly fewer recurrences in boys with prior UTI	Reasonable

- **High-risk boys for whom circumcision is proposed:** recurrent febrile UTI, obstructive uropathy, high-grade hydronephrosis, hydroureteronephrosis, high-grade VUR.

Bladder and Bowel Dysfunction

- **Assess for BBD in every child with UTI**; bladder dysfunction predisposes to **recurrent UTI and renal injury**.
- **Incomplete emptying** (dysfunctional voiding, underactive bladder) raises colonisation and UTI risk.
- **BBD and VUR: voiding against a closed sphincter raises bladder pressure** and may cause and perpetuate VUR; **treating overactive bladder improves spontaneous VUR resolution** — suggesting a causal role.
- **Bowel and bladder are linked:** constipation accompanies 30–88% of bladder dysfunction → **BBD / dysfunctional elimination syndrome (DES)**.

- **VUR plus BBD = highest risk of recurrent pyelonephritis:** recurrence about **45% versus 15%** without BBD.
- **Treating constipation significantly reduces recurrent UTI** and improves bladder function. Detail in Chapter 47.
- **Treatment ladder: timed voiding and constipation treatment** → if failing, directed therapy: pharmacotherapy, biofeedback, electrical stimulation, surgery, CIC, or combinations.
- **Neurogenic bladder on CIC: increasing catheterisation frequency reduces recurrent UTI.**

Management of Vesicoureteral Reflux

- **VUR is a risk factor for recurrent pyelonephritis and scarring;** risk rises with grade and becomes significant with **dilating VUR (grade ≥III).**
- **The dogma that every child with febrile UTI and VUR benefits from diagnosis and treatment** (continuous prophylaxis or surgery) **has been challenged;** management is increasingly complex — detail in Chapter 41.

Concept — *The most effective non-antibiotic prevention is **correcting the host factor** — emptying, constipation, the foreskin — because those, rather than the organism, determine who recurs.*

SEQUELAE OF PAEDIATRIC URINARY TRACT INFECTIONS

Renal Scarring

- **Pyelonephritic scars favour the renal poles,** where **compound papillae** permit intrarenal reflux.
- Risk rises with VUR grade.
- **Imaging: parenchymal thinning over the calyces;** calyces blunted and deformed.
- **Scar versus dysplasia** is hard or impossible to distinguish; a **small kidney with diffusely decreased uptake and low differential function** is usually considered **dysplasia.**
- **Prompt treatment and preventing further pyelonephritis** reduce permanent damage.
- **Delay matters:** odds of new scarring rise with **every hour** of delay and **double when antibiotics start after 72 hours** rather than within 48 hours.
- **Severity gradient:** scar risk rises from **APN → AFBN → renal abscess.**
- **Steroids:** no consistent benefit → **not recommended routinely;** **vitamin A** may reduce scars in girls with APN — investigational.

Xanthogranulomatous Pyelonephritis

- **Xanthogranulomatous pyelonephritis (XGP):** rare chronic inflammatory renal disease in children, usually with **obstruction** and ***Proteus* or *E. coli*.**
- **Presentation:** vague — fever, abdominal or flank pain or **mass,** weight loss, **failure to thrive;** laboratory — leucocytosis, anaemia, pyuria.
- **Imaging:** mass lesion, focal or diffuse, with **perinephric extension** → **mimics malignancy;** CT — poorly enhancing low-attenuation areas and dilated calyces; **calculi in 38–70%.**
- **Treatment:** diffuse → **nephrectomy;** focal → **partial nephrectomy or medical therapy.**
- **Surgery is difficult:** the process encases **psoas and even the great vessels,** destroying planes.

Long-Term Sequelae

- **Hypertension, pre-eclampsia and chronic kidney disease (CKD)** are hard to predict after childhood UTI.
- **Management was built on preventing recurrence and finding VUR;** recent work questions the **UTI/ VUR → scar → hypertension → end-stage disease** chain and implicates **congenital dysplasia.**

- **Registry and literature review:** recurrent childhood UTI accounts for **less than 1% of CKD**; **a child with normal kidneys is not at significant CKD risk** from recurrent UTI.
- **Weak causal link**; risk of end-stage kidney disease after a childhood UTI about **1 in 10,000**.
- **Most children with febrile UTI have no long-term sequelae** (systematic review), especially with **normal kidneys at birth**; decline in function is rare — consistent with the **International Reflux Study in Children** and long adult follow-up.
- **Bilateral scars** lower glomerular filtration rate (GFR) over decades; **unilateral scars do not**; mean ambulatory blood pressure was not different in that cohort.
- **Hypertension is more prevalent in scarred patients** on long-term follow-up and **rises with scar severity**; hypertension in **first pregnancies** is commoner with **severe scarring**.
- **Follow-up:** children with **significant bilateral scars or reduced function** need long-term monitoring of **blood pressure, renal function and proteinuria**; **proteinuria may hasten CKD**; **renin-angiotensin blockade may slow progression**.
- **Large historical cohort:** adolescents with a history of congenital kidney and urinary tract anomalies, pyelonephritis or glomerular disease but **normal adolescent function and no proteinuria** still carried about a **fourfold hazard of later end-stage renal disease**.

KEY POINTS: MANAGEMENT AND SEQUELAE

- **Delay in treating febrile UTI** increases parenchymal involvement and scarring.
- **Hospitalise newborns and young infants** — urosepsis is commoner.
- **Febrile UTI 7–14 days**; **afebrile cystitis 2–4 days**.
- Empirical choice from **local antibiograms**.
- **Nitrofurantoin must not be used for febrile UTI or pyelonephritis**.
- **10–30%** have at least one recurrence.
- Prophylaxis reduces recurrence in **high-risk groups such as girls with dilating reflux**.
- **BBD** raises recurrence risk and should be evaluated and treated in every child with UTI.
- Long-term risk is low, but **bilateral scars or reduced function** warrant long-term follow-up of blood pressure, renal function and proteinuria.

Concept — Long-term harm comes from **the kidney the child was born with and from delayed treatment**, not from UTI as such; so the priorities are prompt treatment and follow-up of the child with bilateral scars or dysplasia.

UNCOMMON PAEDIATRIC URINARY TRACT INFECTIONS

Viral Cystitis

- **Acute haemorrhagic cystitis** is occasionally due to **adenovirus type 11** (also type 21); *E. coli* is found in some, but **most have no identifiable cause**.
- **Common after bone marrow transplantation** and in other immunosuppressed children; untreated severe viral haemorrhagic cystitis carries **50–80% mortality** in these children.
- **BK virus** (DNA polyomavirus): frequent in urine of immunosuppressed and **bone marrow transplant** patients; **90% of adults seropositive** after asymptomatic childhood infection; **latent in the kidney**, reactivates with immune impairment → viraemia.
- **Treatment:** intravenous and intravesical **cidofovir**.

Funguria

- **Increasing**; linked to **recent antibiotics and indwelling catheters**; a **tenfold rise over a decade** in one neonatal intensive care unit.
- **Predisposing factors**: antibiotics, **prematurity**, intravenous and **umbilical artery catheters**, **parenteral nutrition**, immunocompromise.
- **Urinary tract is both portal and site of dissemination**; in disseminated candidiasis **the kidney is the organ most often involved**.
- **Candida is the commonest cause**: *C. albicans* first, then *C. glabrata* (formerly *Torulopsis glabrata*) — often **fluconazole-resistant**.
- **Fungal bezoars** in the renal pelvis can **obstruct** → **ultrasound if funguria persists**.
- **Fungus ball management**: urinary alkalinisation + oral antifungal (may dissolve) → if obstructed, **percutaneous or surgical removal**; percutaneous drainage also allows **local antifungal** (amphotericin B local or systemic, and/or oral fluconazole) → if persistent, **endoscopic or open removal**.
- **Catheter-related asymptomatic funguria**: progression to candidaemia is **rare**; timing of treatment debated; **treat when repeated cultures exceed 10,000–15,000 CFU/mL**.
- **Adjuncts**: stop antibiotics, change or remove catheters, alkalinise urine — helpful but not always curative; **bladder irrigation with amphotericin B** and **oral fluconazole** both clear funguria.
- **Fluconazole** has been used successfully in children, but the chapter lists it as **contraindicated under 6 months**.